STUDENT ACADEMIC MANAGEMENT SYSTEM

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1. Abstract

- The **Student Academic Management System (SAMS)** is a software application designed to streamline and automate the management of academic data within educational institutions. This system aims to replace traditional paper-based or manual methods by providing a centralized digital platform to manage student records, subjects, attendance, grades, and academic performance efficiently and accurately.

2. Introduction

- **1.Definition:** A Student/Academic Management System is a digital tool that automates and manages academic and administrative tasks in educational institutions.
- **2.Users:** The system is used by administrators, teachers, students, and sometimes parents for seamless communication and data handling.
- **3.Features:** Common features include student registration, timetable scheduling, grade tracking, attendance monitoring, examination management, and reporting tools.

- **4.Benefits:**Reduces paperwork and manual errors

  Enhances communication between stakeholders

  Saves time and administrative effort

  Allows quick access to academic records
- **5. Technology**: Can be implemented as web-based or mobile applications, often using modern technologies like cloud storage, databases, and user authentication systems.
- **6.Goal:** To digitize academic operations and support data-driven decision-making in educational institutions.

3.Justification

The Student Academic Management System simplifies and automates the management of student data, subjects, and marks. It reduces manual errors, saves time, and provides quick access to academic records. By using Python and SQLite, the system is lightweight, cost-effective, and ideal for small institutions. It also helps track student performance efficiently and can be expanded with additional features in the future.

3. Problem Statement

- Inefficiency and errors in manual record keeping
- Difficulties in managing large student databases
- Lack of quick access to academic data
- There is a lack of centralized systems to store and manage student records such as personal details, grades, attendance, and course information.
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4. Objectives

- Efficient Student Information Management
- Streamlined Academic Processes
- Attendance and Discipline Tracking
- Communication and Collaboration
- Resource Management
- Reporting and Analytics
- User Access Control
- Integration and Scalability

5. System Requirements

- **Functional Requirements**:These describe what the system should do:User ManagementRegister/login for students, teachers, and adminsRole-based access controlProfile management
- Student Management:Add, edit, delete student recordsTrack attendance and gradesView academic progress reports
- Course & Class Management:Create/manage courses and subjectsAssign teachers to coursesSchedule classes and exams
- Enrollment Management:Student registration for coursesView course catalogPrerequisite and eligibility checks
- Reports & Analytics:Academic performance dashboardsAttendance and fee reportsExport data (CSV, PDF, etc.)

- **Non-Functional Requirements**:These describe how the system should perform:PerformanceFast response times(e.g.,<2sfor most queries)Support concurrent users (scale based on institution size)
- Security:Role-based permissionsData encryption (especially for sensitive student data)Regular backups and audit trails
- Scalability:Scalable to handle more students, courses, and campuses
- Usability: Intuitive UI/UX for all rolesAccessible on mobile and desktop devices
- Maintainability: Modular codebase for easy updates Well-documented API and backend
- Availability: 99.9% uptime with failover mechanismsScheduled maintenance notifications

6.Methodology

- **Methodology:**The development of the Student/Academic Management System followed a structured software development lifecycle to ensure a robust and user-friendly application. The methodology is divided into the following phases
- 1. Requirement Analysis:This phase involved gathering and analyzing the needs of end-users including administrators, faculty, and students.
- 2.System Design:A detailed design was created using UML diagrams, ER diagrams (for database structure), and wireframes for the user interface. This helped in visualizing the system architecture and data flow between modules such as:

Student Registration **Attendance Tracking** **Timetable Scheduling**

Course Management **Grade Management**

- 3. Development: The system was developed using [insert technologies here, e.g., Python/Django, PHP/Laravel, Java/Spring Boot, etc.] for the backend, and HTML/CSS/JavaScript (with or without frameworks) for the frontend. A relational database (e.g., MySQL or PostgreSQL) was used to store academic data securely.
- 4. Testing: The system underwent multiple levels of testing:Unit Testing for individual componentsIntegration Testing to ensure modules work togetherUser Acceptance Testing (UAT) with actual users to gather feedback
- 5. Evaluation: Effectiveness was measured by user satisfaction surveys and comparing manual processes vs. automated processes in terms of time efficiency and accuracy.

7.ER Diagram

- 1. Student

  StudentID (PK), FirstName, LastName , DOB , Email ,Phone ,Address, EnrollmentDate

  2. Course

  CourseID (PK), CourseName, CourseDescription ,Credits

  3. Instructor

  InstructorID (PK), FirstName, LastName, Email, Department

  4. Department

  DepartmentID(PK), DepartmentName, OfficeLocation

5. Enrollment

EnrollmentID(PK), StudentID(FK), CourseID(FK), EnrollmentDate, Grade

6. ClassSchedule

ScheduleID(PK), CourseID(FK), InstructorID(FK), Semester, Time, Location

Relationships:

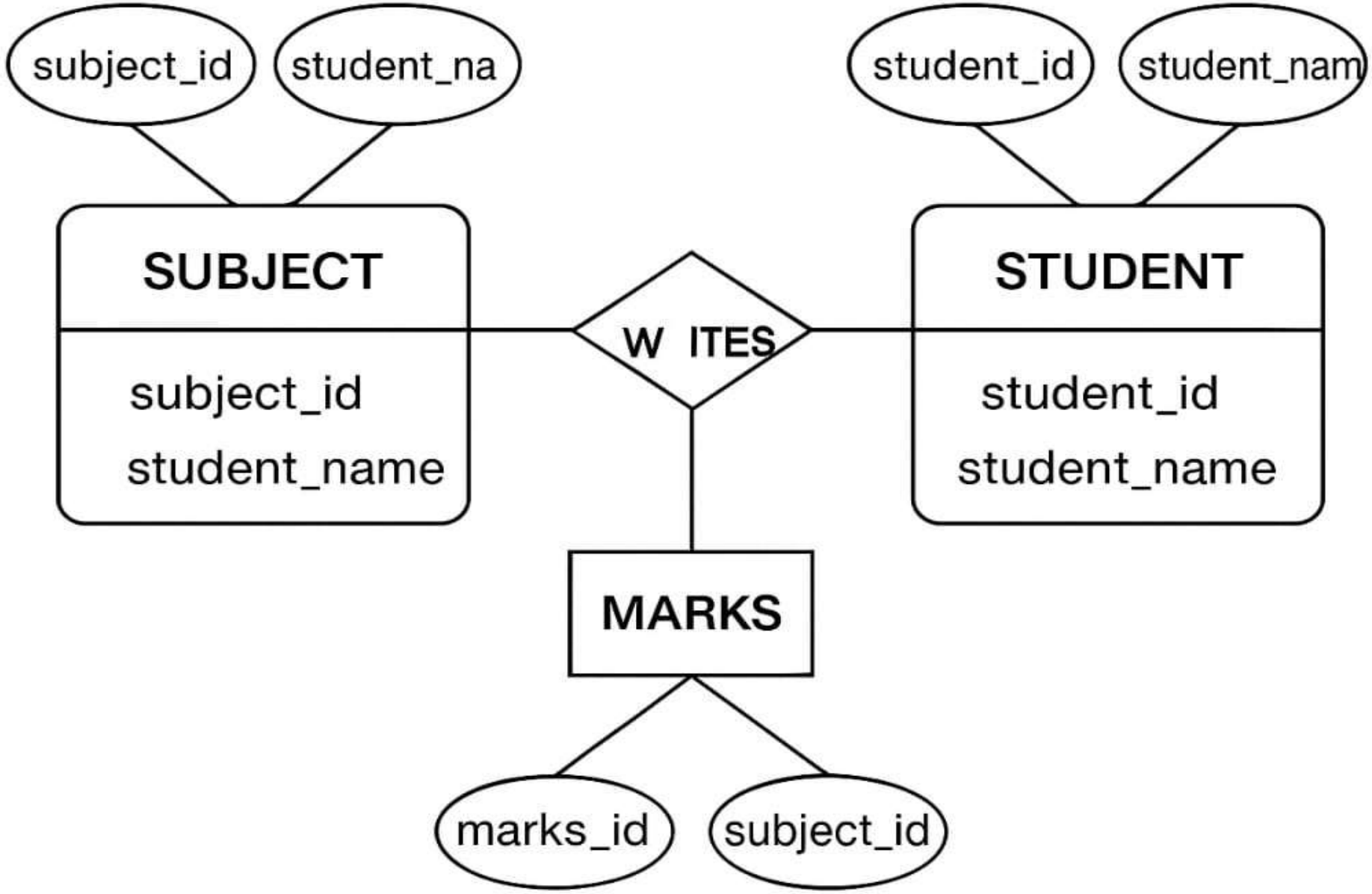
A Student can enroll in many Courses (Many-to-Many via Enrollment).

A Course can have many Students (Many-to-Many via Enrollment).

A Course is taught by one or more Instructors (Many-to-Many via ClassSchedule).

An Instructor belongs to one Department (Many-to-One).

A Department can have many Instructors (One-to-Many).


subject_id
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SUBJECT
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W ITES
STUDENT
student_id
student_name
MARKS
marks_id
subject_id

```
import sqlite3

# Connect s)
conn = sqlite3.connect('student_manement.db')
cursor = conn.cursor()

# Create tables
cursor.execute('''
CREATE TABLE IF NOT EXISTS students (
    student_id INTEGER PRIMARY KEY AUTOINCREMENT,
    name TEXT NOT NULL
)
''')

cursor.execute('''
CREATE TABLE IF NOT EXISTS subjects (    subject_id INTEGER PRIMARY KEY AUTOINCREMENT,
    name TEXT NOT NULL
)
''')
```

```
cursor.execute('''
CREATE TABLE IF NOT EXISTS marks (
    mark_id INTEGER PRIMARY KEY AUTOINCREMENT,
    student_id INTEGER,
    subject_id INTEGER,
    marks INTEGER,
    FOREIGN KEY(student_id) REFERENCES students(student_id),
    FOREIGN KEY(subject_id) REFERENCES subjects(subject_id)
)
''')
# Function to add a student
def add_student(name):
    cursor.execute("INSERT INTO students (name) VALUES (?)", (name,))
    conn.commit()
    print(f"Student '{name}' added.")
# Function to add a subject
def add_subject(name):
    cursor.execute("INSERT INTO subjects (name) VALUES (?)", (name,))
    conn.commit()
    print(f"Subject '{name}' added.")
# Function to record marks
def record_marks(student_id, subject_id, marks):
```

```
cursor.execute("INSERT INTO marks (student_id, subject_id, marks) VALUES (?,
?, ?)",
                (student_id, subject_id, marks))
    conn.commit()
    print(f"Marks recorded for Student ID {student_id} in Subject ID {subject_id}:
{marks} marks.")
# Function to display student performance
def show_student_performance(student_id):
    cursor.execute('''
    SELECT students.name, subjects.name, marks.marks
    FROM marks
    JOIN students ON marks.student_id = students.student_id
WHERE students.student_id = ?
    ''', (student_id,))
    records = cursor.fetchall()
    if not records:
        print("No records found.")
        return
 print(f"\nPerformance for Student ID {student_id} ({records[0][0]}):")
    for subject, marks in [(r[1], r[2]) for r in records]:
```

```
print(f"- {subject}: {marks} marks")
# Sample usage

if __name__ == "__main__":
    add_student("Alice")
    add_student("Bob")

    add_subject("Math")
    add_subject("Science")

    record_marks(1, 1, 85)  # Alice - Math
    record_marks(1, 2, 92)  # Alice - Science
    record_marks(2, 1, 78)  # Bob - Math

    show_student_performance(1)
    show_student_performance(2)

    conn.close()
```

Student 'Alice' added.
Student 'Bob' added.
Subject 'Math' added.
Subject 'Science' added.
Marks recorded for Student ID 1 in Subject ID 1: 85 marks.
Marks recorded for Student ID 1 in Subject ID 2: 92 marks.
Marks recorded for Student ID 2 in Subject ID 1: 78 marks.

Performance for Student ID 1 (Alice):
- Math: 85 marks
- Science: 92 marks

Performance for Student ID 2 (Bob):
- Math: 78 marks

10.Conclusion

- The Student/Academic Management System is a vital tool for modern educational institutions,enabling efficient handling of student data, academic records,attendance,and examination results.By automating routine tasks and centralizing information,the system improves administrative efficiency,enhances accuracy,and supports better decision-making….

11.REFERENCES

- **Moodle** – Open-source Learning Management System (LMS).

  URL: https://moodle.org

- **EDUCAUSE** – https://www.educause.edu/

  A leading resource on IT in higher education including SIS (Student Information Systems).

THANK YOU